蔡承耘 Tsai Cheng Yun

jimmytsai1017@gmail.com | +886 921-340-452 | Taichung, Taiwan

Military Status: Completed

Education

National Taiwan University (NTU) 2023 – 2025
M.S. in Construction Management, Civil Engineering *GPA: 4.1 / 4.3*

- Courses: Applied Deep Learning, Practical Data Structures and Algorithms, Computer Vision, Database Management System –from SQL to NoSQL (Cumulative GPA: 4.075)
- Award: Dean’s Award

National Taiwan University (NTU) 2019 – 2023
B.S. in Civil Engineering *GPA: 3.7 / 4.3*

- Award: 111-1 Semester Outstanding Teaching Assistant Award

Experience

Stony Brook University, Visiting Scholar Feb – Aug 2025

- Developed and evaluated a multi-person, multi-camera localization and re-identification system for factory environments using a Unity-based digital twin and open-source computer vision frameworks.

Stantec Consulting Services Inc., Summer Intern Jun – Aug 2024

- Developed a prototype Retrieval-Augmented Generation (RAG) system using LangChain, Gemini, and VectorDB to support internal regulation and case retrieval.

NTUCE-NCREE AI Research Center, Summer Intern Jun – Aug 2022

- Benchmarked RGB-D camera and LiDAR in SLAM tasks using ROS and RTAB-Map, and developed a sensor fusion pipeline achieving an effective range of 20m —4× that of RGB-D alone —while retaining RGB texture and IMU odometry.

Publication

Multi-granular Crew Activity Recognition for Construction Monitoring Nov 2025
Published in Automation in Construction [ScienceDirect]

- Propose a multi-granular activity recognition framework for construction crew monitoring.
- Model hierarchical relationships between crew- and individual-level activities.

Projects

Evaluate Consistency between Multi-modal Engineering Design Documents Apr 2025

- Developed a multi-modal pipeline combining YOLO-based object detection, and VLM-based visual semantic understanding to automatically extract and validate consistency across drawings and documents.

Global Motion Compensation system [GitHub] May 2024

- Implemented a global motion compensation pipeline using feature matching and homography estimation to stabilize video sequences under camera motion.

Floor Plan Automatic Generation May 2023

- Developed a low-cost floor plan generation pipeline by capturing iPhone video and applying Structure-from-Motion (SfM) via COLMAP to reconstruct dense 3D point clouds, followed by RANSAC and Hough Transform for wall extraction and 2D floor plan generation.

Skills

Programming: Python, C, C++
Tools: Unity, Revit, AutoCAD, SketchUp, COLMAP
Frameworks: PyTorch, OpenCV
Language: English (TOEIC: 880)